

Business Analysis Report

Integrated Customer Analysis — Python, SQL & Power BI

1. Executive Summary

- The analysis is built around a 50,000-row, 53-column customer dataset. The Python notebook verifies unique customer IDs and performs missing-value treatment, exploratory analysis, customer value/RFM analysis, churn-risk analysis, category profitability/returns analysis, and shopping-channel analysis.
- Customer value is strongly represented by the Very High customer-value category and High CLV category in the Python visuals. SQL further provides formal ranking logic for top customers by spending, CLV and profitability, plus segment-level CLV/profitability analysis.
- Churn/inactivity is measurable through churn-risk category, recency, activity status, engagement and health metrics. The Python churn chart shows Low and Very Low risk as the largest groups, while High and Very High risk groups are smaller but directly relevant for retention action.
- Health & Personal Care is the category with the highest average customer-level return rate in the Python return-rate chart. This makes it the clearest return-related investigation priority supported by the available visual evidence.
- The uploaded materials do not contain a monthly revenue series or a cohort-retention dataset. Therefore, the questions about why revenue fell last month and which cohort has best retention cannot be answered reliably from the supplied evidence.
- The SQL script contains strong analytical logic—aggregations, subqueries, CTEs, joins and window functions—but it contains query definitions rather than saved query-result tables. Consequently, several exact winners/rates cannot be reported without inventing values.
- The Power BI file contains four report pages: Executive Overview, Customer Intelligence, Product and Profitability, and Marketing Intelligence. Importantly, the Executive Overview KPI queries are hard-filtered to preferred_category_1 = 'Books' in the PBIX layout, so those cards should not be interpreted as unrestricted company-wide KPIs.

2. Business Objective

The purpose of this analysis is to turn the customer dataset into management-ready evidence about revenue, customer value, inactivity/churn, profitability, returns, marketing-channel quality and retention. The requested 10 questions are decision-oriented, but the supplied evidence is primarily a customer-level analytical dataset rather than a transaction-level time series. The report therefore distinguishes established findings from questions that require additional data.

3. Data Sources & Analysis Tools

Python Analysis

The notebook loads the customer dataset into pandas, verifies customer-ID uniqueness, inspects columns/shape/types, checks missing values and duplicates, fills selected categorical missing values, performs EDA and creates visuals for customer demographics, value/CLV, RFM, churn risk, category spending/profit, returns and marketing behaviour. The notebook also calculates return_rate and exports a cleaned CSV. No machine-learning model is present in the supplied notebook.

Python evidence	Observed detail
Dataset size	50,000 rows × 53 columns
Customer ID	Unique according to the notebook check
Missing-value treatment	preferred_category_2 → None; preferred_category_3 → None; loyalty_tier → Unknown; social_media_presence → None
Duplicate check	0 duplicate rows reported
Return-rate calculation	return_count / total_purchases × 100, rounded to 2 decimals
Return-rate missing values	13 were filled with 0 before later dropping return_rate for export

SQL Analysis

The SQL script uses the table `customer_analysis_cleaned_one`. It covers customer counts, segment/gender/country/loyalty aggregation, top-10 spending/CLV/profitability, segment spending and CLV, CLV/RFM/churn categories, category purchases, shopping-channel CLV, CLV classification, subqueries, joins, CTEs and window functions.

Technique	Evidence in SQL
Aggregations	COUNT, SUM, AVG, GROUP BY, ORDER BY
Join	Customer rows joined to segment-level average CLV
CTE	Segment/CLV analysis and ranking queries
Subqueries	Overall-average spending/CLV comparisons
Window functions	RANK, DENSE_RANK, ROW_NUMBER, LAG, running SUM
Business calculations	Spending contribution %, running total, category rankings, segment profitability/CLV

Power BI Dashboard

The PBIX contains four report pages: Executive Overview, Customer Intelligence, Product and Profitability, and Marketing Intelligence. Across these pages the model exposes KPI cards and visuals for revenue, profit, CLV, return rate, average order value, customers, repeat customer rate, RFM, customer type, category revenue/spending/profitability, profit margin, returns, acquisition cost, conversion, channel revenue and channel customer counts.

Page	Key visual evidence
Executive Overview	Total Revenue, Total Profit, Avg CLV, Return Rate, Avg Order Value, Total Customers; slicers for age group, category, segment, channel and country
Customer Intelligence	Repeat Customer Rate by customer segment; customer counts; RFM category; customer type; CLV by segment; Top Customers
Product and Profitability	Profitability by category; Total Revenue; category spending; Profit Margin %; Return Rate; Top/Bottom Products
Marketing Intelligence	Acquisition cost, channel return rate, profitability/cost, revenue, conversion rate and channel customer count

PBIX caveat: the Executive Overview KPI queries in the uploaded layout include a fixed filter `preferred_category_1 = 'Books'`. Therefore those KPI cards are not safe to interpret as unrestricted company-wide totals without opening the report and removing/confirming that filter.

4. Data Preparation & Methodology

Analytical workflow: Raw Customer Data → Python Cleaning & EDA → SQL Business Analysis → Power BI Visualization → Business Interpretation.

Python cleaning

Loaded the CSV, checked uniqueness, schema, shape, data types, missingness and duplicates; filled selected categorical nulls; derived `return_rate`; inspected customer, value, churn, category, return and channel patterns.

SQL business analysis

Used a cleaned customer table to aggregate, rank and compare customers, segments, categories and channels; added business logic with CTEs, subqueries and window functions.

Power BI presentation

Organized the analysis into four management-facing pages with KPI cards, comparison visuals and slicers.

Decision support

Integrated what is measurable across sources and explicitly flagged questions for which the uploaded files do not contain the required time-series/cohort/result evidence.

5. Detailed Business Question Analysis

Q1. Why did revenue decrease last month?

Status: NOT DETERMINABLE

Business Question

Why did revenue decrease last month?

Analysis

The supplied Python notebook is a customer snapshot with a dataset_year field, but it does not show a monthly revenue series. The SQL script has no date/month revenue query, and the Power BI visual inventory has no monthly revenue trend visual. Therefore there is no evidence for a last-month decline, its magnitude, or its drivers.

Key Metrics / Evidence

Required metric	Availability in supplied files
Monthly revenue trend	Not available
Last month vs prior month	Not available
Revenue decline amount/%	Not available
Product/channel/segment drivers	Not attributable without monthly revenue

Finding

The supplied Python notebook is a customer snapshot with a dataset_year field, but it does not show a monthly revenue series. The SQL script has no date/month revenue query, and the Power BI visual inventory has no monthly revenue trend visual. Therefore there is no evidence for a last-month decline, its magnitude, or its drivers.

Business Interpretation

Management should not attribute the decline to a product, channel or customer factor from this evidence alone. A transaction/order-level date field and monthly revenue table are required.

Supporting Evidence

No rendered source visual in the uploaded artifacts directly answers this question; the limitation is recorded rather than replaced with a synthetic chart.

Business Implication

Management should not attribute the decline to a product, channel or customer factor from this evidence alone. A transaction/order-level date field and monthly revenue table are required.

Q2. Which customers are most valuable?

Status: SUPPORTED — GROUP LEVEL

Business Question

Which customers are most valuable?

Analysis

The Python analysis includes total spending, customer profitability, purchase frequency, RFM score/category, customer value category and CLV category. The Very High customer-value category is the largest value group in the Python chart, while High CLV is the dominant CLV category. SQL explicitly ranks customers by total spending, CLV and profitability and also ranks top CLV customers within each segment.

Key Metrics / Evidence

Value lens	Evidence
Customer value category	Very High is the largest category in the Python visual
CLV category	High CLV is the dominant category in the Python visual
SQL ranking	Top 10 by spending, CLV and profitability are queried
Segment ranking	Average CLV and total profitability are queried by customer segment

Finding

The Python analysis includes total spending, customer profitability, purchase frequency, RFM score/category, customer value category and CLV category. The Very High customer-value category is the largest value group in the Python chart, while High CLV is the dominant CLV category. SQL explicitly ranks customers by total spending, CLV and profitability and also ranks top CLV customers within each segment.

Business Interpretation

The strongest supported definition of valuable customers is high CLV / high value, reinforced by spending and profitability. Exact customer IDs and values are not reproduced because the SQL file contains the ranking queries but not their result rows.

Supporting Evidence

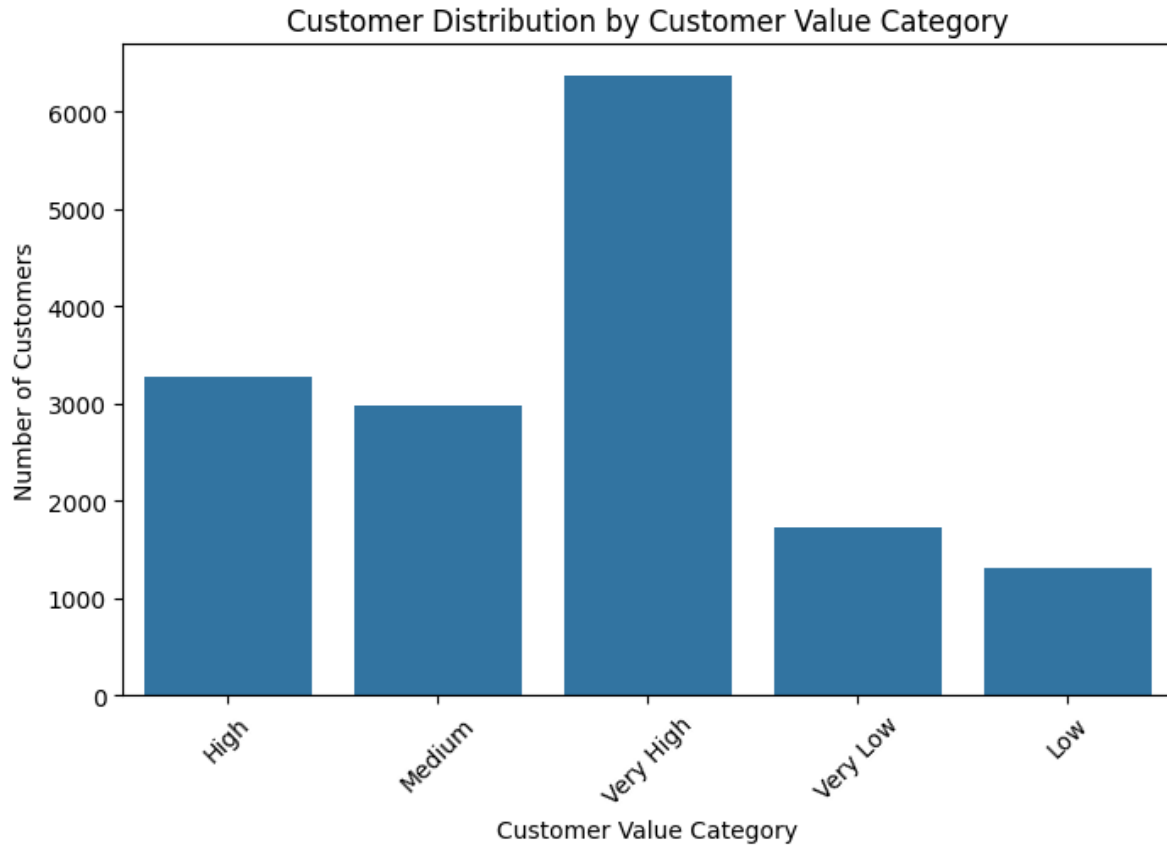


Figure 1. Customer distribution by customer value category — Python notebook.

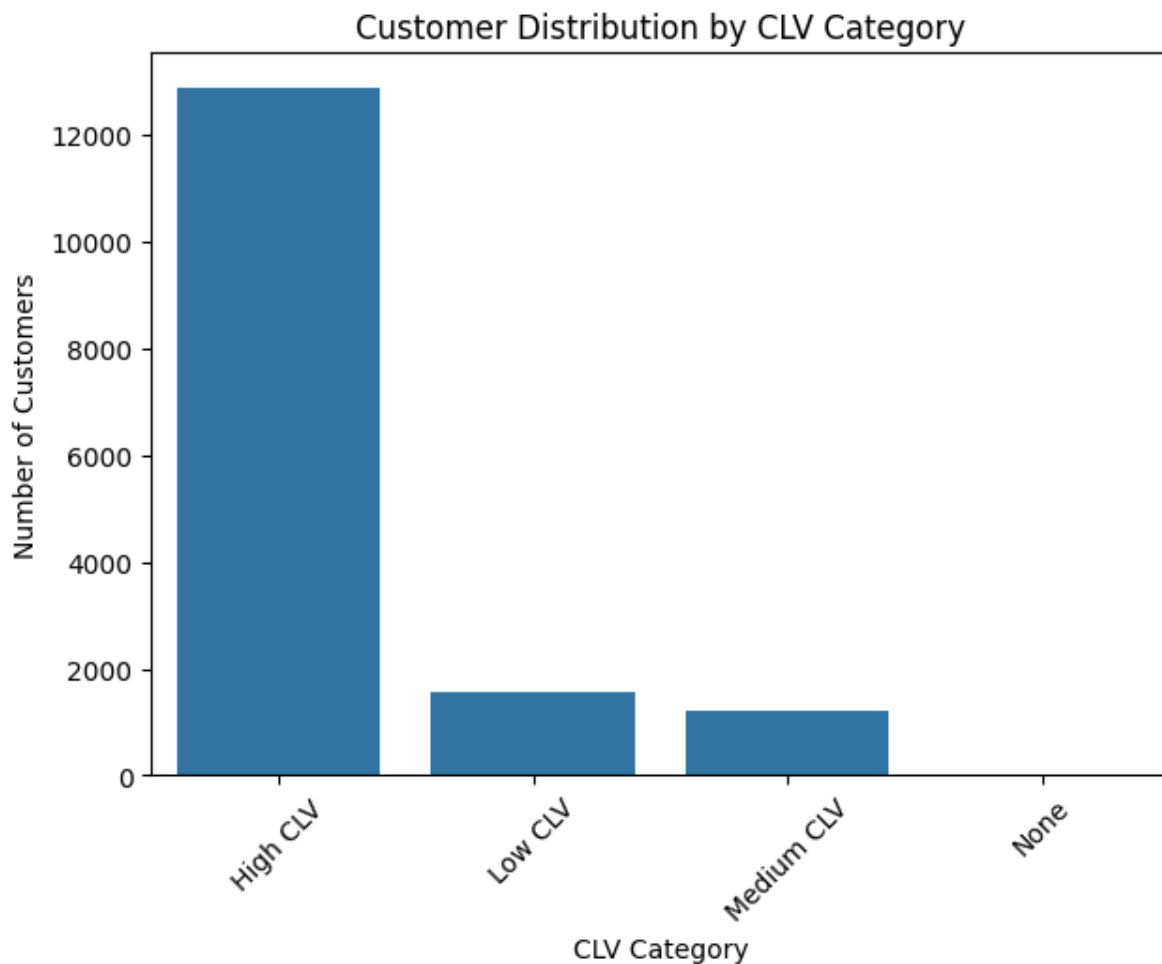


Figure 2. Customer distribution by CLV category — Python notebook.

Business Implication

The strongest supported definition of valuable customers is high CLV / high value, reinforced by spending and profitability. Exact customer IDs and values are not reproduced because the SQL file contains the ranking queries but not their result rows.

Q3. Which customers are at risk of becoming inactive?

Status: SUPPORTED — RISK GROUPS

Business Question

Which customers are at risk of becoming inactive?

Analysis

The Python notebook directly analyzes churn_risk_score, churn_risk_category, activity_status, recency_score, engagement_level, customer_health_score and health_status. The churn-risk visual shows Low and Very Low as the largest groups, with smaller High and Very High groups. The dataset also includes days_since_last_purchase and RFM fields.

Key Metrics / Evidence

Risk lens	Evidence
Churn risk	High and Very High groups are visible in the Python distribution
Recency	days_since_last_purchase and recency_score are available
Activity	activity_status is analyzed
RFM/health	rfm_category, engagement_level, customer_health_score and health_status are analyzed

Finding

The Python notebook directly analyzes churn_risk_score, churn_risk_category, activity_status, recency_score, engagement_level, customer_health_score and health_status. The churn-risk visual shows Low and Very Low as the largest groups, with smaller High and Very High groups. The dataset also includes days_since_last_purchase and RFM fields.

Business Interpretation

The High/Very High churn-risk groups are the clearest retention-priority population. The evidence supports targeted investigation, but the files do not provide a saved customer-level risk list.

Supporting Evidence

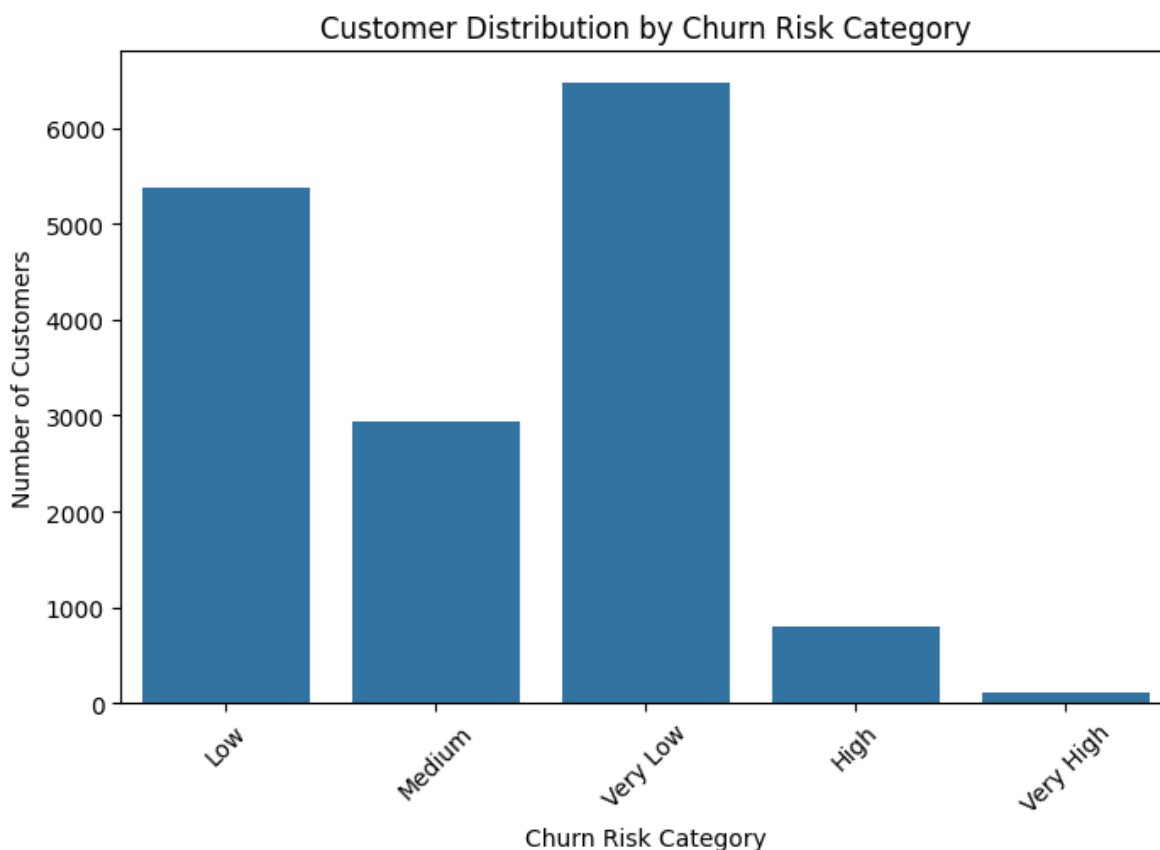


Figure 3. Customer distribution by churn-risk category — Python notebook.

Business Implication

The High/Very High churn-risk groups are the clearest retention-priority population. The evidence supports targeted investigation, but the files do not provide a saved customer-level risk list.

Q4. Which products generate revenue but not enough profit?

Status: PARTIALLY SUPPORTED

Business Question

Which products generate revenue but not enough profit?

Analysis

The dataset is category-oriented through preferred_category_1 rather than a product-SKU field in the Python notebook. Python compares total spending and total customer profitability by category. The Product and Profitability Power BI page contains Profit Margin %, Top Products and Bottom Products visuals, but the PBIX does not expose their rendered values in the uploaded artifact.

Key Metrics / Evidence

Evidence	Conclusion
Product SKU field	Not present in the notebook's 53-column schema
Category spending	Available
Category customer profitability	Available
Profit margin	Available as a Power BI measure, but rendered values are not exposed in the uploaded artifact

Finding

The dataset is category-oriented through preferred_category_1 rather than a product-SKU field in the Python notebook. Python compares total spending and total customer profitability by category. The Product and Profitability Power BI page contains Profit Margin %, Top Products and Bottom Products visuals, but the PBIX does not expose their rendered values in the uploaded artifact.

Business Interpretation

Toys has the lowest total customer profitability among the plotted categories, but it also has relatively low spending, so it cannot be called a strong-revenue/weak-profit product from the evidence. A defensible product-level answer requires the rendered margin/product values or a product-level source table.

Supporting Evidence

No rendered source visual in the uploaded artifacts directly answers this question; the limitation is recorded rather than replaced with a synthetic chart.

Business Implication

Toys has the lowest total customer profitability among the plotted categories, but it also has relatively low spending, so it cannot be called a strong-revenue/weak-profit product from the evidence. A defensible product-level answer requires the rendered margin/product values or a product-level source table.

Q5. Which category has the highest return problem?

Status: SUPPORTED

Business Question

Which category has the highest return problem?

Analysis

Python calculates return_rate as $\text{return_count} / \text{total_purchases} \times 100$ and fills 13 undefined values with zero. The category return-rate chart shows Health & Personal Care with the highest average plotted return rate.

Key Metrics / Evidence

Metric	Evidence
Return rate formula	$\text{return_count} / \text{total_purchases} \times 100$
Highest plotted category	Health & Personal Care
Undefined return rates	13 filled with 0
Revenue impact of returns	Not calculated in supplied Python/SQL results

Finding

Python calculates return_rate as $\text{return_count} / \text{total_purchases} \times 100$ and fills 13 undefined values with zero. The category return-rate chart shows Health & Personal Care with the highest average plotted return rate.

Business Interpretation

Health & Personal Care is the first category to investigate for return drivers. The analysis should distinguish high return frequency from revenue impact because the supplied notebook does not calculate category revenue lost to returns.

Supporting Evidence

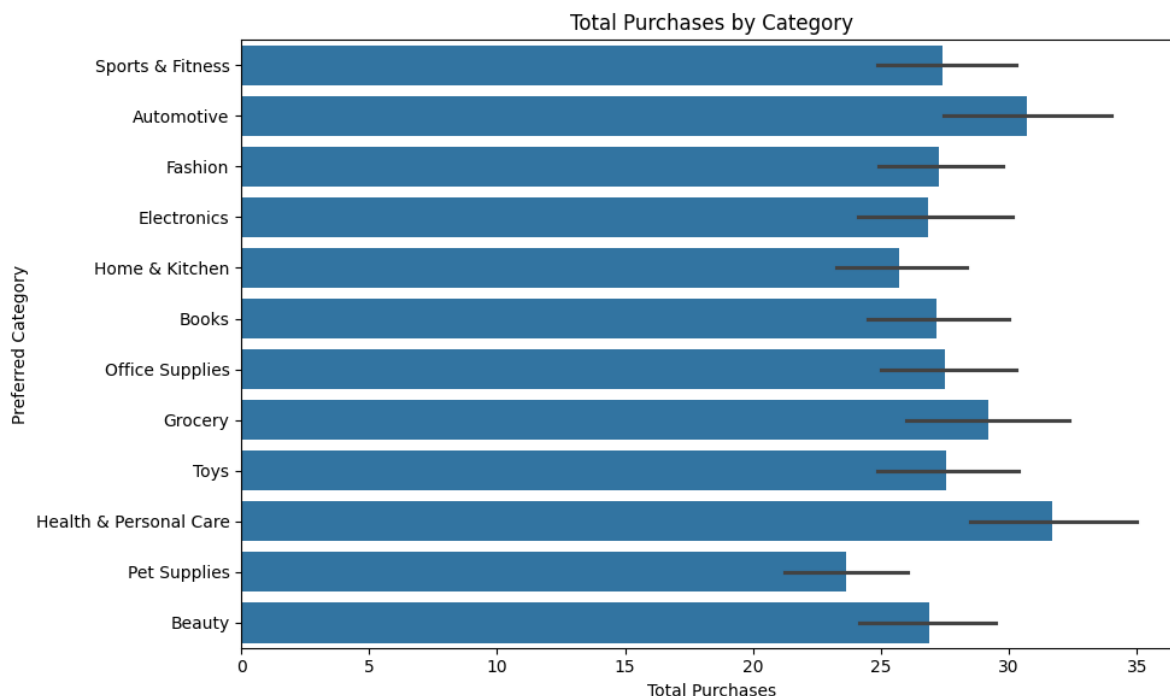


Figure 4. Average plotted return rate by preferred category — Python notebook.

Business Implication

Health & Personal Care is the first category to investigate for return drivers. The analysis should distinguish high return frequency from revenue impact because the supplied notebook does not calculate category revenue lost to returns.

Q6. Which marketing channel brings the highest-value customers?

Status: NOT DETERMINABLE — RESULT MISSING

Business Question

Which marketing channel brings the highest-value customers?

Analysis

SQL explicitly asks for the shopping channel with the highest average CLV. The Power BI Marketing Intelligence page compares channels using acquisition cost, return rate, profitability/cost, revenue, conversion rate and customer count. However, the SQL result is not included and the Power BI page does not expose a channel-by-CLV value in the uploaded layout.

Key Metrics / Evidence

Channel evidence	Availability
Channel customer distribution	Available in Python
Channel revenue	Available in Power BI
Channel conversion	Available in Power BI
Channel acquisition cost	Available in Python/Power BI
Channel average CLV winner	SQL query exists; result not supplied

Finding

SQL explicitly asks for the shopping channel with the highest average CLV. The Power BI Marketing Intelligence page compares channels using acquisition cost, return rate, profitability/cost, revenue, conversion rate and customer count. However, the SQL result is not included and the Power BI page does not expose a channel-by-CLV value in the uploaded layout.

Business Interpretation

The Python channel distribution is broadly balanced across In-store, Marketplace, Online and Mobile App, but customer count alone cannot answer the question. The saved evidence is insufficient to name the highest-value channel.

Supporting Evidence



Figure 7. Customer distribution by shopping channel — Python notebook.

Business Implication

The Python channel distribution is broadly balanced across In-store, Marketplace, Online and Mobile App, but customer count alone cannot answer the question. The saved evidence is insufficient to name the highest-value channel.

Q7. Which customer cohort has the best retention?

Status: NOT DETERMINABLE

Business Question

Which customer cohort has the best retention?

Analysis

The uploaded Python notebook does not create cohort labels or a cohort-retention matrix. The Power BI Customer Intelligence page contains Repeat Customer Rate by customer segment, not a cohort-retention visual. No cohort field or saved retention table is present in the supplied evidence.

Key Metrics / Evidence

Required cohort evidence	Availability
Cohort label	Not present
Cohort retention matrix	Not present
Retention by cohort	Not available
Repeat Customer Rate by segment	Present in Power BI, but not a cohort analysis

Finding

The uploaded Python notebook does not create cohort labels or a cohort-retention matrix. The Power BI Customer Intelligence page contains Repeat Customer Rate by customer segment, not a cohort-retention visual. No cohort field or saved retention table is present in the supplied evidence.

Business Interpretation

A cohort definition based on first purchase period and a cohort-by-month retention matrix are required before selecting a best-retaining cohort.

Supporting Evidence

No rendered source visual in the uploaded artifacts directly answers this question; the limitation is recorded rather than replaced with a synthetic chart.

Business Implication

A cohort definition based on first purchase period and a cohort-by-month retention matrix are required before selecting a best-retaining cohort.

Q8. What is our repeat purchase rate?

Status: PARTIALLY SUPPORTED — VALUE MISSING

Business Question

What is our repeat purchase rate?

Analysis

The Power BI Customer Intelligence page contains a 'Repeat Customer Rate' measure plotted by customer segment. The SQL file, however, does not define or calculate a repeat-purchase-rate formula, and the uploaded layout does not reveal the measure's underlying DAX definition or rendered value.

Key Metrics / Evidence

Repeat-purchase evidence	Availability
Repeat Customer Rate measure	Present in Power BI
Measure definition/DAX	Not exposed in uploaded layout
Rendered rate	Not available
SQL calculation	Not present

Finding

The Power BI Customer Intelligence page contains a 'Repeat Customer Rate' measure plotted by customer segment. The SQL file, however, does not define or calculate a repeat-purchase-rate formula, and the uploaded layout does not reveal the measure's underlying DAX definition or rendered value.

Business Interpretation

The existence of the metric is validated, but its exact rate and formula cannot be responsibly reported from the uploaded files alone.

Supporting Evidence

No rendered source visual in the uploaded artifacts directly answers this question; the limitation is recorded rather than replaced with a synthetic chart.

Business Implication

The existence of the metric is validated, but its exact rate and formula cannot be responsibly reported from the uploaded files alone.

Q9. Where are we losing revenue?

Status: SUPPORTED AS RISK AREAS; LOSS NOT QUANTIFIED

Business Question

Where are we losing revenue?

Analysis

The evidence points to four measurable investigation areas: returns (especially Health & Personal Care), customer inactivity/churn risk, category profitability differences, and marketing-channel quality. The SQL also calculates spending contribution, running total, segment profitability/CLV and category rankings. However, no transaction-level revenue-loss bridge is supplied.

Key Metrics / Evidence

Revenue-leakage area	Evidence level
Returns	Measured rate; not monetized
Churn/inactivity	Measured risk categories
Profitability	Category/customer profitability available
Channel quality	Multiple channel metrics available
Dollar revenue loss	Not quantified

Finding

The evidence points to four measurable investigation areas: returns (especially Health & Personal Care), customer inactivity/churn risk, category profitability differences, and marketing-channel quality. The SQL also calculates spending contribution, running total, segment profitability/CLV and category rankings. However, no transaction-level revenue-loss bridge is supplied.

Business Interpretation

The report can identify revenue-leakage risk areas, but it cannot claim a measured dollar loss for any area. Returns, churn and low repeat behavior should be treated as potential leakage mechanisms until quantified against transaction-level revenue.

Supporting Evidence

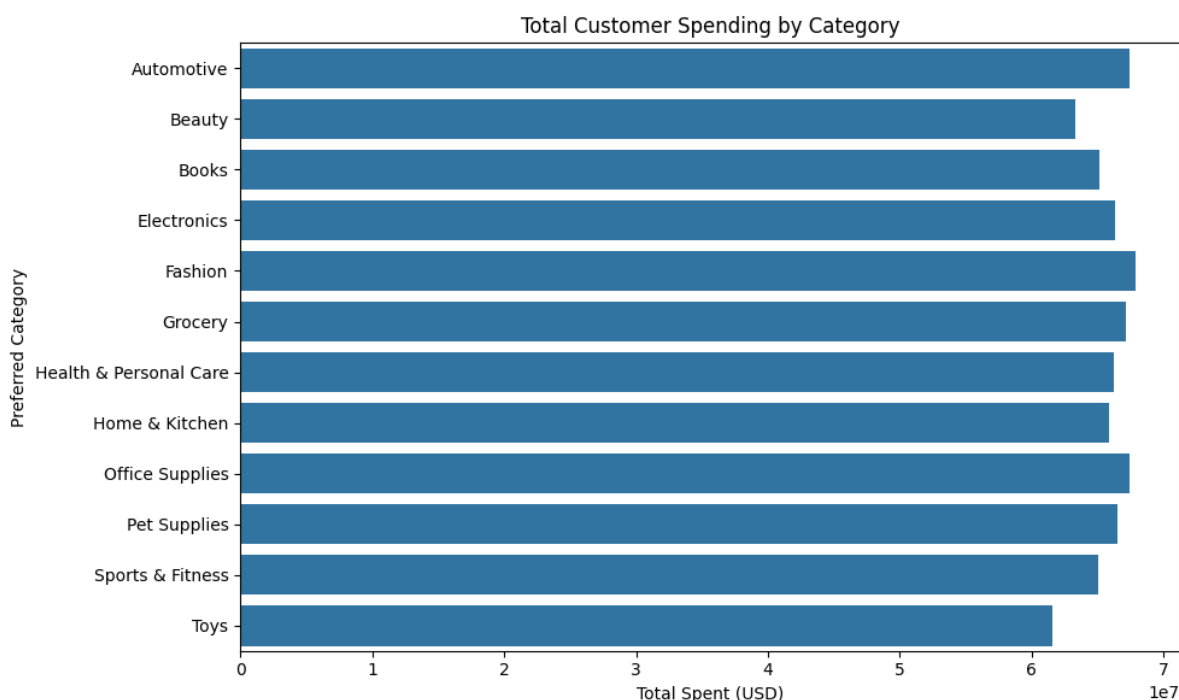


Figure 5. Total customer spending by category — Python notebook.



Figure 6. Total customer profitability by category — Python notebook.

Business Implication

The report can identify revenue-leakage risk areas, but it cannot claim a measured dollar loss for any area. Returns, churn and low repeat behavior should be treated as potential leakage mechanisms until quantified against transaction-level revenue.

Q10. What should management investigate next?

Status: SUPPORTED — PRIORITIZED

Business Question

What should management investigate next?

Analysis

The next investigations should follow the strongest evidence: return drivers in Health & Personal Care; High/Very High churn-risk customers; high-value customer retention; category profit-margin differences; and channel quality using CLV and profitability rather than customer count alone.

Key Metrics / Evidence

Priority	Reason
1	Return investigation in Health & Personal Care
2	High/Very High churn-risk retention
3	High-value customer protection
4	Channel quality by CLV/profitability
5	Add monthly revenue and cohort-retention data

Finding

The next investigations should follow the strongest evidence: return drivers in Health & Personal Care; High/Very High churn-risk customers; high-value customer retention; category profit-margin differences; and channel quality using CLV and profitability rather than customer count alone.

Business Interpretation

These actions are directly linked to the observed Python, SQL and Power BI structures and do not assume an unsupported revenue uplift.

Supporting Evidence

No rendered source visual in the uploaded artifacts directly answers this question; the limitation is recorded rather than replaced with a synthetic chart.

Business Implication

These actions are directly linked to the observed Python, SQL and Power BI structures and do not assume an unsupported revenue uplift.

6. Screenshot & Visual Evidence

The report embeds actual chart outputs preserved in the uploaded Python notebook. The PBIX was inspected as a Power BI report package; its four page definitions and visual queries were reviewed. Because the uploaded PBIX does not contain static page screenshots, no synthetic Power BI screenshot has been substituted. The report instead documents the actual PBIX page/visual structure and flags where rendered values are unavailable.

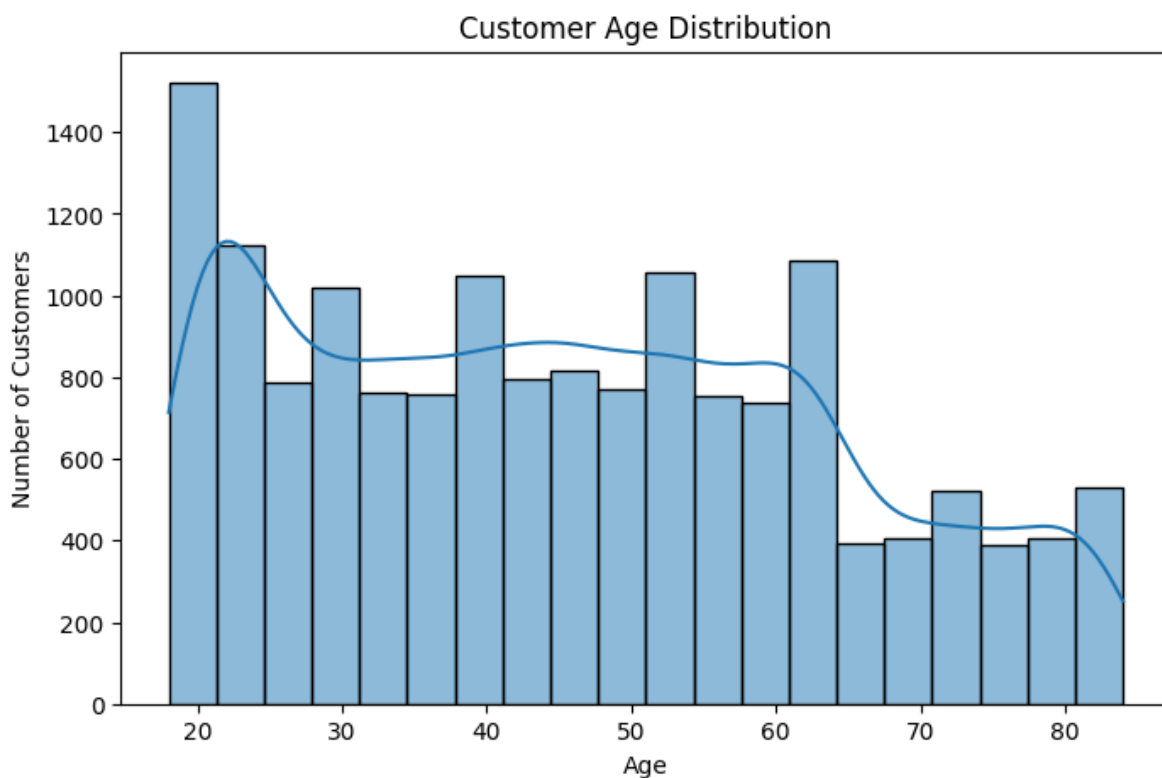


Figure 8. Customer age distribution — Python notebook.

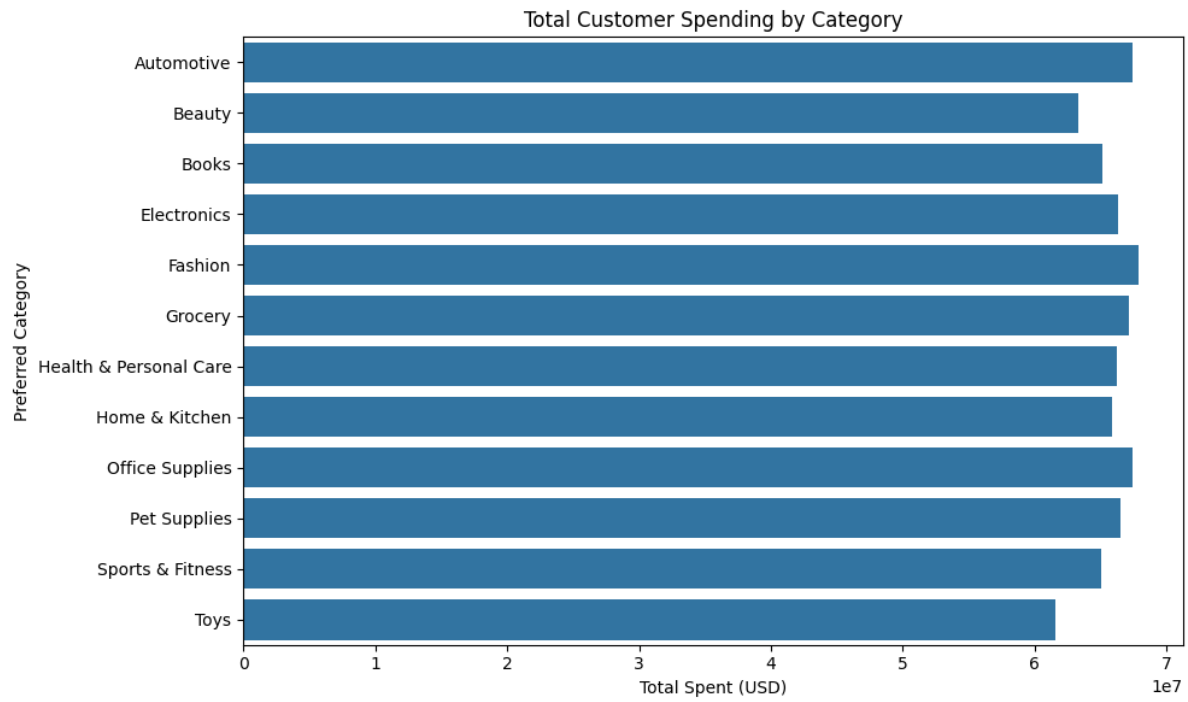


Figure 9. Total customer spending by preferred category — Python notebook.



Figure 10. Customer distribution by shopping channel — Python notebook.



E-Commerce Customer 360° And Revenue Intelligence Analytics Executive Overview

207.22M

Total Revenue



237.40M

Total Profit



4K

Total Customers



10

Return Rate



501

Avg of order value



57K

Avg of CLV



Customer Intelligence

Product and Profitability

Marketing Intelligence

Show Slicers

Fig 11.PowerBI Dashboard Executive Overview

7. Cross-Source Validation

Finding	Python	SQL	Power BI	Validation
Customer value	Value/CLV distributions	Top-10 and segment ranking queries	CLV by segment + Top Customers	Consistent analytical framing; exact SQL results unavailable
Churn/inactivity	Churn, recency, activity, health	Churn-risk category count query	Customer Intelligence includes customer analytics	Consistent dimensions; no saved customer-risk output
Returns	Return rate by category	Return-risk category query not present; fields are available	Return Rate KPI and category return visual	Strong metric alignment; revenue impact not quantified
Marketing channels	Channel distribution + engagement metrics	Average-CLV-by-channel query	Revenue, conversion, CAC, return rate, profitability by channel	Channel quality is analyzable, but highest-CLV result is missing
Retention	No cohort analysis	No repeat-rate SQL	Repeat Customer Rate by segment	Repeat metric exists; cohort retention does not

8. KPI Summary

KPI / measure	Evidence status	Interpretation
Customer count	50,000 rows in Python dataset	Dataset-level customer population
Unique customer IDs	Yes	No duplicate customer IDs indicated by uniqueness check
Total Revenue	Available in Power BI; Executive Overview is Books-filtered	Do not treat as company-wide without confirming/removing filter
Total Profit	Available in Power BI; Executive Overview is Books-filtered	Same filter caveat
Avg CLV	Available in Power BI; Books-filtered on Executive Overview	CLV is a central value metric
Return Rate	Available in Power BI and calculated in Python	Health & Personal Care has highest plotted average return rate
Repeat Customer Rate	Available as Power BI measure	Rendered value/definition not available in

		uploaded artifact
Profit Margin %	Available as Power BI measure	Rendered category/channel values not available
Revenue Growth/Decline	Not available as monthly time series	Cannot answer last-month decline question

9. Key Business Insights

Revenue & Profitability

- Monthly revenue decline cannot be established from the supplied customer-level snapshot.
- Category spending and profitability are available, but a true revenue-loss bridge requires transaction-level dates and revenue components.
- The Power BI Product and Profitability page contains Profit Margin %, but the uploaded artifact does not expose its rendered values.

Customer Value

- Very High is the largest customer-value category in the Python visual.
- High CLV is the dominant CLV category in the Python visual.
- SQL supports top-customer ranking by spending, CLV and profitability and top-3 CLV ranking within each segment.

Customer Churn & Inactivity

- Low and Very Low churn-risk groups dominate the Python distribution.
- High and Very High risk groups are smaller but are the most relevant population for proactive retention analysis.
- Recency, activity status, engagement and health fields provide multiple lenses for prioritization.

Product Performance

- The available source is category-oriented rather than SKU-oriented.
- Toys shows the lowest total customer profitability in the plotted category comparison, but also lower spending; this is not sufficient to label it a high-revenue/low-profit product.

Returns

- Health & Personal Care has the highest average plotted return rate.
- The notebook also tracks complaint count and satisfaction score, providing supporting dimensions for a return investigation.

Marketing Performance

- Customer counts are relatively balanced across In-store, Marketplace, Online and Mobile App in the Python visual.
- The dashboard evaluates channels with acquisition cost, return rate, profitability, revenue, conversion and customer count.
- Highest-value channel cannot be named because the SQL CLV query has no saved result.

Customer Retention

- Power BI includes Repeat Customer Rate by customer segment.
- No cohort definition or cohort-retention matrix is present in the supplied evidence.

Revenue Leakage

- Return frequency, churn risk, category profitability and channel quality are the strongest observable leakage-risk areas.
- None is converted into a verified dollar revenue-loss figure in the supplied files.

10. Management Recommendations

Recommendation 1

Finding: Investigate Health & Personal Care returns

Business Issue: The Python return-rate visual places Health & Personal Care highest.

Recommended Action: Review return reasons, product mix, customer complaints and revenue associated with returned purchases.

Expected Impact: Expected impact is not quantified in the supplied files.

Data Evidence: Python return-rate analysis; Product and Profitability Power BI page.

Recommendation 2

Finding: Create a retention watchlist for High/Very High churn risk

Business Issue: The churn-risk chart contains identifiable High and Very High groups.

Recommended Action: Join churn risk with CLV, recency, spending and profitability to prioritize outreach.

Expected Impact: Potential value preservation is plausible, but no uplift is estimated in the files.

Data Evidence: Python churn-risk, recency, CLV and profitability fields; SQL customer ranking logic.

Recommendation 3

Finding: Protect high-value customers

Business Issue: Very High customer-value and High CLV groups are prominent in the Python visuals.

Recommended Action: Segment high-value customers by loyalty, RFM, channel and activity status and monitor changes over time.

Expected Impact: No monetary uplift should be assumed without an intervention test.

Data Evidence: Python customer-value/CLV visuals; SQL CLV ranking and segment analysis.

Recommendation 4

Finding: Validate channel quality using value, not volume

Business Issue: Channel customer counts are broadly similar, while SQL explicitly seeks the channel with highest average CLV.

Recommended Action: Produce a channel scorecard with average CLV, profitability, revenue, conversion and acquisition cost.

Expected Impact: The supplied evidence does not quantify the business impact of reallocating spend.

Data Evidence: SQL channel-CLV query; Power BI Marketing Intelligence visuals.

Recommendation 5

Finding: Add a monthly revenue bridge

Business Issue: The current files cannot explain a 'last month' decline.

Recommended Action: Add transaction-level dates and create month-over-month revenue, orders, AOV, returns and channel/segment decomposition.

Expected Impact: Impact cannot be quantified until the missing time series is available.

Data Evidence: Gap identified across Python, SQL and Power BI.

Recommendation 6

Finding: Build a cohort-retention layer

Business Issue: No cohort retention dataset is present, although repeat customer rate exists in Power BI.

Recommended Action: Add first-purchase cohort, month-since-acquisition and retention-rate calculations.

Expected Impact: No retention uplift is estimated from the supplied evidence.

Data Evidence: Power BI Repeat Customer Rate visual; missing cohort fields identified during source review.

11. What Should Management Investigate Next?

What to investigate	Why	Evidence	Suggested next step
Revenue decline	Question cannot be answered without time series	No monthly revenue field/visual	Add transaction date and build month-over-month revenue bridge
Low-profit products/categories	Margin evidence exists but rendered values are missing	Power BI Product and Profitability page	Export/render category/product margin values and compare with revenue
High-return category	Health & Personal Care has highest plotted return rate	Python return-rate visual	Break returns down by product, reason, complaints and revenue affected
High-value retention	Very High value and High CLV groups are prominent	Python value/CLV visuals; SQL rankings	Build a high-value retention/watchlist view
At-risk customers	High/Very High churn-risk groups exist	Python churn-risk visual	Prioritize by CLV, recency, spending and profitability
Marketing channel quality	Channel counts alone are insufficient	SQL CLV query + Power BI channel metrics	Publish average CLV and profitability by channel
Repeat purchase behaviour	Repeat Customer Rate exists but is not quantified	Power BI Customer Intelligence	Expose measure definition/value and add one-time vs repeat counts
Cohort retention	No cohort data is present	Source review	Add first-purchase cohort and retention matrix

12. Final Conclusion

The integrated evidence shows a customer analytics solution with strong coverage of customer value, CLV, RFM, churn risk, category performance, returns and marketing-channel dimensions. The strongest immediately supported findings are the prominence of Very High customer-value/High CLV groups, the existence of a smaller High/Very High churn-risk population, and the elevated return-rate pattern for Health & Personal Care. SQL adds a robust business-analysis layer through rankings, segment comparisons, subqueries, CTEs and window functions, while Power BI packages the analysis into four management-oriented pages.

At the same time, the evidence has important boundaries: it is not sufficient to prove a month-over-month revenue decline, identify a best-retaining cohort, quantify repeat purchase rate, or name the highest-CLV marketing channel from the supplied artifacts alone. Management should therefore treat the current dashboard as a strong customer-analysis foundation and add transaction dates, cohort fields and saved result outputs before making claims about revenue decline, retention rates or monetized leakage.